

Algebra 1
Unit 4
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

1. Complete the table of values for $y = -\frac{3}{4}x - 3$.

$$-\frac{3}{4}(4) - 3 = -3 - 3 = -6$$

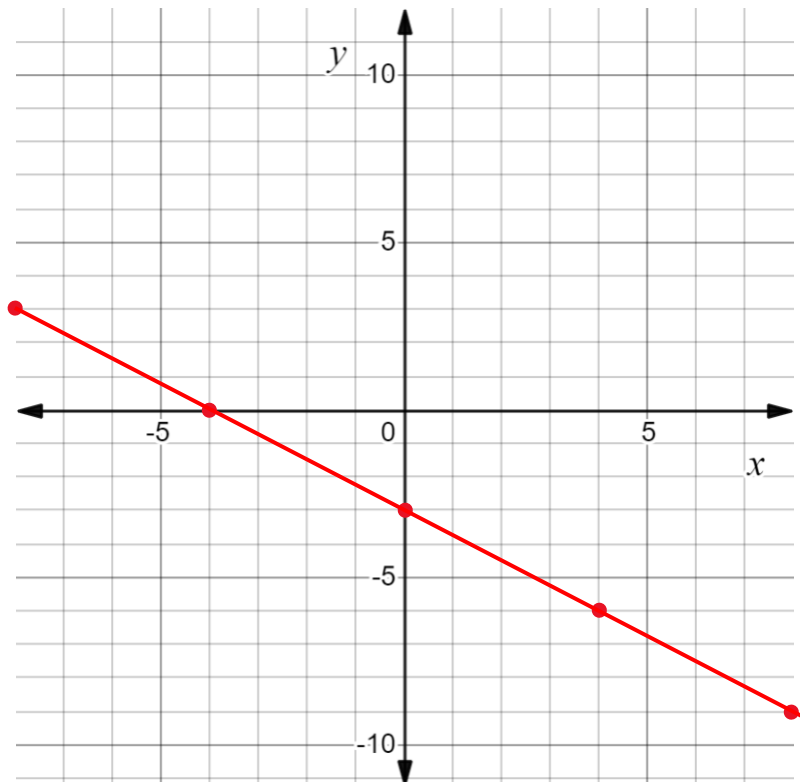
x	-8	-4	0	4	8
y	3	0	-3	-6	-9

$$-\frac{3}{4}(-8) - 3 = 6 - 3 = 3$$

$$-\frac{3}{4}(-4) - 3 = 3 - 3 = 0$$

$$-\frac{3}{4}(8) - 3 = -6 - 3 = -9$$

2. Graph the equation $y = -\frac{3}{4}x - 3$.



3. Write the domain of the line $y = -\frac{3}{4}x - 3$ using set-builder notation.

$$\{x: -\infty < x < \infty\}$$

4. What is the x -intercept of the line $y = -\frac{3}{4}x - 3$?

$$(-4, 0)$$

5. Complete the statement about the line $y = -\frac{3}{4}x - 3$.

As x -values increase, y -values decrease

6. Complete the table of values for $y + 5 = -4(x - 2)$.

$$y + 5 = -4(-1 - 2) \rightarrow y + 5 = -4(-3) \rightarrow y = 12 - 5 = 7$$

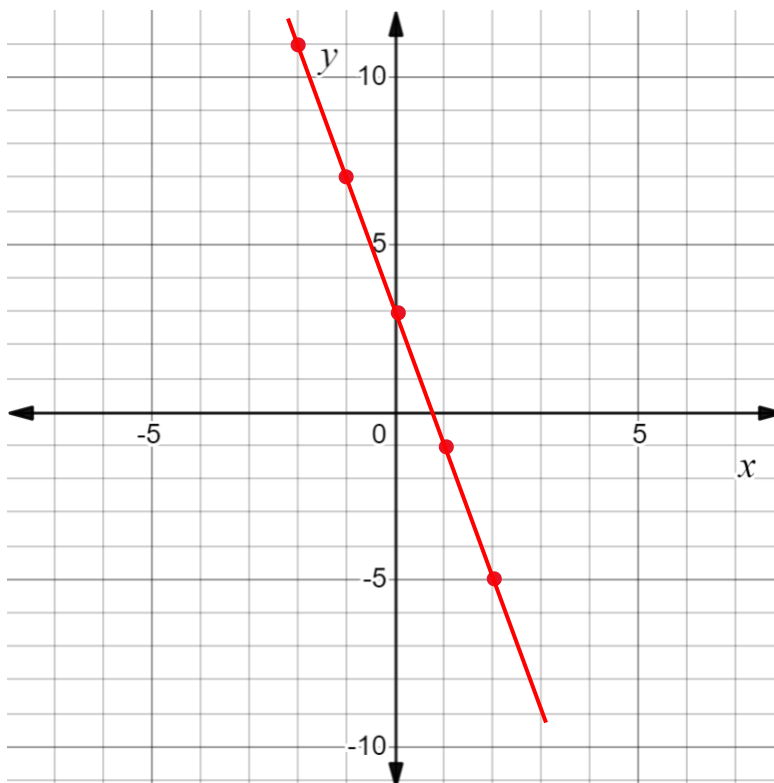
$$y + 5 = -4(-1) \rightarrow y = 4 - 5 = -1$$

x	-2	-1	0	1	2
y	11	7	3	-1	-5

$$y + 5 = -4(-2 - 2) \rightarrow y + 5 = -4(-4) \rightarrow y = 16 - 5 = 11$$

$$y + 5 = -4(-2) \rightarrow y = 8 - 5 = 3$$

7. Graph the equation $y + 5 = -4(x - 2)$.



8. Use inequalities to complete the table for the line $y + 5 = -4(x - 2)$.

Domain	Range
$-\infty < x < \infty$	$-\infty < y < \infty$

9. What is the y -intercept of the line $y + 5 = -4(x - 2)$?

(0,3)

10. Complete the statement about the line $y - 3 = 3(x + 1)$.

As x -values increase, y -values increase.